EVERY FINITE SEPARABLE RESIDUE EXTENSION IS REALIZED BY AN ETALE NEIGHBOURHOOD

THE STACKS PROJECT AUTHORS

Editorial note. Statement span is lines 10137-10145 of sources/stacks-more-morphisms/pinned-original-source.tex; proof span is lines 10147-10151, the complete author proof (3 body lines).

Editorial note. Navigation stubs: 1 cited-but-not-included labels are supplied as non-load-bearing pointers with their pinned tags where the snapshot tag index has them.

Editorial note. extra_packages is empty on purpose: the retained preamble already loads xy with options and a second declaration clashes in the assembler wrapper.

Editorial note. Text is retained verbatim, including the author's own typos and placeholders.

Editorial note. The reduction 'we may assume S affine' is written in the span; the card is not a bare pointer.

Allowed standard prerequisites (editorial).

- Explicit prerequisite boundary. The algebraic construction of an etale ring map with prescribed residue field extension (Algebra, lemma-make-etale-map-prescribed-residue-field); cited input with the affinization reduction written in the retained span.

Source reference. Original lemma 00UD: sources/stacks-more-morphisms/pinned-original-source.tex, source lines 10151–10151. Cited by the retained statement or proof but not part of the collected spans; supplied only as a navigation pointer. The argument does not depend on its content.

Lemma 0.1. *Let S be a scheme. Let $s \in S$. Let $k/\kappa(s)$ be a finite separable field extension. Then there exists an étale neighbourhood $(U, u) \to (S, s)$ such that the field extension $\kappa(u)/\kappa(s)$ is isomorphic to $k/\kappa(s)$.*

Proof. We may assume S is affine. In this case the lemma follows from Algebra, Lemma 00UD. □

Date: stacks-project:more-morphisms.tex:local-snapshot-20261011; GNU FDL 1.2 or later.